

DATA SHEET for RISC-V based I3C master

Register Map

CAPABILITIES AND OPERATION REGISTERS:

Register Name	Address offset	Description
Interrupt status register	0x20	The Interrupt Status register reflects the status of outstanding interrupt(s). Bit 0:ahb_wr_almost_empty Bit 1:ahb_wr_empty Bit 2:ahb_wr_almost_full Bit 3:ahb_wr_full Bit 4:comm_fifo_empty Bit 5:comm_fifo_full Bit 6:ahb_rd_full Bit 7:ahb_rd_empty
Interrupt status enable register	0x24	Enables the corresponding interrupt bits when its set to 1'b1
Device address section offset	BASE+0x30	It holds the size and offset of DAT table
Device address table(0-32)	0x30+(0x00 ,0x04)	DAT table stores the address, information and control fields of device attached to I3C bus
Device characteristics section offset register	BASE+0x34	This DCT table offset section holds the size and offset of the table
Device characteristics table (0-64)	0x34+(0x00,0x04)	DCT table captures the device characteristics (PID,BCR,DCR) and assigned dynamic address for the device attached to I3C. <i>Note : Each device contains 4 DCT table.</i>

Externally defined registers

Register Name	Address offset	Description
Command transfer register0	0x0C	This register indicates the type of command

		format which I3C has to execute
Command transfer register1	0x10	Second half of command transferring registers
Broadcast address register	0x18	This register has the broadcast address of I3C bus
Clock port register	0x44	This register is for clock frequency control in I3C bus in both open drain and push pull mode
Ideal time register	0x78	It holds the Ideal time count before starting an I3C transaction.
Write data port register	0x84	This has the write data need to be written in I3C data line during write transaction from master
Read data port register	0x88	This has the data had been read from I3C during read transaction

Detailed description for commands and operational Registers:

Note: For more information refer MIPI_I3C_HCI specification

Command transfer for REGULAR DATA TRANSFER

Bits	Field name	Memory access	Reset value	Description
32+(31:16)	Data length	W	0x0	Data transfer data length Indicates the number of bytes to be transferred
(32+)15:0	Reserved	-	-	-
31	TOC	W	0x0	Data transfer terminate on completion 1'b0-repeated start after data transfer 1'b1- stop after data transfer
30	ROC	W	0x0	Data transfer response on completion Controls whether response status is required after successful compilation of the transfer command.
29	RNW	W	0x0	Data transfer r/w 0-write

				▪ 1-read
28:26	Mode	W	0x0	Data transfer speed and mode ▪ 0x0-sdr0-12.5MHZ ▪ 0x1-sdr1-8MHZ ▪ 0x2-sdr2-6MHZ ▪ 0x3-sdr3-4MHZ ▪ 0x4-sdr4-2MHZ ▪ 0x5-HDR_TS ▪ 0x6-HDR_DDR
25:21	RESERVED	-	-	-
20:16	DEV_INDEX	W	0x0	Data transfer device index Static and device addressing related information will be stored to this index in the DAT.
15	CP	W	0x0	Data transfer command present ▪ 0-SDR transfer ▪ 1-HDR transfer
14:7	CMD	W	0x0	Data transfer CCC ▪ For CCC:8bits ▪ For HDR:7 bits
6:3	TID	W	0x0	Data transfer transaction ID
2:0	CMD_ATTR	W	0x0	Command type defining the format of the other fields. ▪ 0x0 for regular data transfer

Command transfer register for IMMEDIATE DATA TRANSFER

Bits	Field name	Memory access	Reset value	Description
32+(31:24)	Data_byte_4	W	0x0	Immediate data transfer data byte4
32+(23:16)	Data_byte_3	W	0x0	Immediate data transfer data byte3
32+(15:8)	Data_byte_2	W	0x0	Immediate data transfer data byte2
32+(7:0)	Data_byte_1	W	0x0	Immediate data transfer data byte1
31	TOC	W	0x0	Data transfer terminate on completion 1'b0-repeated start after data transfer 1'b1- stop after data transfer
30	ROC	W	0x0	Data transfer response on completion Controls whether response status is required after successful compilation of the transfer command.
29	RNW	W	0x0	Data transfer r/w 0-write 1-read
28:26	Mode	W	0x0	Data transfer speed and mode 0x0-sdr0-12.5MHZ 0x1-sdr1-8MHZ 0x2-sdr2-6MHZ 0x3-sdr3-4MHZ 0x4-sdr4-2MHZ 0x5-HDR_TS 0x6-HDR_DDR
25:23	Byte count			Number of valid data bytes to use in this immediate data transfer descriptor
22:21	RESERVED	-	-	-

20:16	DEV_INDEX	W	0x0	Data transfer device index Static and device addressing related information will be stored to this index in the DAT.
15	CP	W	0x0	Data transfer command present ▪ 0-SDR transfer ▪ 1-CCC or HDR transfer
14:7	CMD	W	0x0	Data transfer CCC ▪ For CCC:8bits ▪ For HDR:7 bits
6:3	TID	W	0x0	Data transfer transaction ID
2:0	CMD_ATTR	W	0x0	Command type defining the format of the other fields. ▪ 0x1 for immediate data transfer

Command structure for ADDRESS ASSIGNMENT COMMAND

Bits	Field name	Memory access	Reset value	Description
63:32	Reserved	-	-	-
31	TOC	W	0x0	Data transfer terminate on completion 1'b0-repeated start after data transfer 1'b1- stop after data transfer
30	ROC	W	0x0	Data transfer response on completion Controls whether response status is required after successful compilation of the transfer command.
29:26	DEV_COUNT	W	0x0	Indicates the number of devices that a dynamic address shall be assigned to

25:21	RESERVED	-	-	-
20:16	DEV_INDEX	W	0x0	Data transfer device index Static and device addressing related information will be stored to this index in the DAT.
15	RESERVED	-	-	-
14:7	CMD	W	0x0	Specifies ccc code indicating whether address assignment uses ENTDAAs or SETDASAs commands
6:3	TID	W	0x0	Data transfer transaction ID
2:0	CMD_ATTR	W	0x0	Command type defining the format of the other fields. 0x2 for address assignment command

Command Descriptor for INTERNAL_CONTROL

Bits	Field Name	Memory Access	Reset Value	Description
(32+)31:0	Vendor Specific	W	0x0	Vendor Specific
31:12	MIPI_RESERVED	W	0x0	MIPI Alliance reserved area
11:8	MIPI_CMD	W	0x0	MIPI Alliance command values 0x0: NoOP 0x1: Lock/Unlock 0x2: Include 7E(IBA) 0x3: Device context update
7	VENDOR_INFO_PRESENT	W	0x0	Indicates whether a Vendor-Specific extension is present
6:3	TID	W	0x0	Identification tag for command
2:0	CMD_ATTR	W	0x0	Command attribute 0x7: INTERNAL_CONTROL: Internal control command

Internal control structure, Fields for lock/unlock sub-command

Bits	Field Name	Memory Access	Reset Values	Description
31:13	MIPI reserved	W	0x0	MIPI Alliance eserved area
12	ON_OFF	W	0x0	Bus Instance lock/unlock 1'b0: lock off 1'b1: Lock on

Internal Control Structure, Fields for 0x7E Sub-command

Bits	Field Name	Memory Access	Reset Values	Description
31:13	MIPI_RESERVED	W	0x0	MIPI Alliance reserved area
12	ON_OFF	W	0x0	Bus Instance on/off 1'b0:IBA_INCLUDE off 1'b1: IBA_INCLUDE on

Internal control structure, Fields for Device Context Update Command

Bits	Fields Name	Memory Access	Reset Value	Description
31:19	MIPI_RESERVED	W	0x0	MIPI Alliance reserved area
18:14	INDEX	W	0x0	Indicates the DCT table index for the slave device
13:12	OP_TYPE	W	0x0	Device Context Entry Opertaion Type 0x0: Device Context entry is being added 0x1: Device Context entry is being changed 0x2: Device Context entry is being changd

Response queue port(BASE+ 0x4)

Bits	Field Name	Memory Access	Rest Value	Descriptor
31:28	ERR_STATUS	R	0x0	Response Error Status 0x0: SUCESS: Transfer successful no

				error ▪ 0x1: CRC: CRC Error ▪ 0x2: PARITY: Parity error ▪ 0x3: FRAME: Frame Error ▪ 0x4: ADDR_HEADER: Address Header Error ▪ 0x5: NACK: Address Nack’ed or Dynamic address Nack’ed ▪ 0x6: OVL: Receive overflow or underflow error ▪ 0x7:RESERVED ▪ 0x8: ABORTED: Aborted ▪ 0x9: I2C_WR_DATA_NACK: Nack received for I2C write data transfer ▪ 0xA:NOT_SUPPORTED ▪ 0xB: RESERVED
27:24	TID	R	0x0	Identification tag for the command. This value match one of the command sent on the Bus ▪ 0x0-0x7: Valid Transaction Ids ▪ 0x8-0xF: Reserved
23:16	RESERVED	—	—	—
15:0	DATA_LENGTH	R	0x0	Data length/Device Count ▪ For write transfer: Remaining data length(in bytes) ▪ For Read transfer: Received data length(in bytes) ▪ For address assignment: Remaining Device count

Device address table structure(BASE+0x_n00, BASE +0x_n04)

Bits	Field name	Memory access	Reset value	Description
(32+)	reserved	R/W	0x0	-

31:27				
(32+) 26:19	Autcmd_hdr_code	R/W	0x0	Specifies auto command read command code. Only valid if read is executed in HDR mode.
(32+) 18:16	Autocmd_mode	R/W	0x0	Indicates the mode and speed for the auto commands read
(32+) 15:8	Autocmd_value	R/W	0x0	Value of the IBI mandatory byte that will trigger an auto command read transaction on the i3c bus
(32+) 7:0	Autocmd_mask	R/W	0x0	Mask for the IBI mandatory byte that will trigger auto command read transaction on the i3c bus
31	device	R/W	0x0	Device type values 0x0 i3c device 0x1 i2c device
30:29	Dev_nack_retry_cnt	R/W	0x0	Device nack specific retry count
28:26	Ring_id	R/W	0x0	Device ring group id used to send an IBI from specific device to the appropriate ring bundle.
25:24	reserved	R/W	0x0	-
23:16	Dynamic address	R/W	0x0	Device i3c dynamic address
15	TS	R/W	0x0	Device IBI timestamp
14	MR reject	R/W	0x0	Device In band master interrupt request reject
13	SIR reject	R/W	0x0	Device In band slave interrupt request reject when operating in master role
12	IBI payload	R/W	0x0	Device IBI payload Indicates whether IBIs from this device have a data payload. This field reflects the IBI payload bit in device characteristic register
11:7	reserved	R/W	0x0	-

6:0	Static address	R/W	0x0	Device static address i3c/i2c static address
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Dct_reg(Device characteristic table)(0x_n00,0x_n04,0x_n08,0x_n1C)

Bits	Field name	Memory access	Reset value	Description
(96+) 31:8	Reserved	-	-	-
(96+) 7:0	Dynamic address	R	0x0	Device i3c dynamic address include parity bit
(64+) 31:16	Reserved	-	-	-
(64+) 15:8	BCR	R	0x0	Device bus characteristic register
(64+) 7:0	DCR	R	0x0	Device i3c device characteristic register
(32+) 31:16	Reserved	-	-	-
(32+) 15:0	PID	R	0x0	Device provisional id low
31:0	PID	R	0x0	Device provisional id High

Detailed descriptions for externally defined registers:

Bits	Register name	Memory access	Reset value	Description
31:0	Write data port register	W	0x0	DATA write register which has the data to be sent from I3C transaction
31:0	Read data port register	R	0x0	DATA read register which has the value read by I3C during read transaction.

31:0	Clock port register	W	0x0	Clock port register give the values of push pull and open drain mode clock divisions [7:0]->push pull clock division value *2 for division by 4 [15:8]->open drain clock division value *16 for division by 32 <i>Note: Give value as 1002 in hex value</i>
31:0	Command transfer register0	W	0x0	First half of command transfer values which is totally 64 bits, Immediate command, Regular command, Address assignment command, Internal control command.
31:0	Command transfer register1	W	0x0	Second half of command transfer values, Immediate command, Regular command, Address assignment command, Internal control command.
31:0	Ideal time register	W	0x0	Ideal count time value which has to be maintained during I3C transaction
31:0	Broadcast address register	W	0x0	This register would have the broadcast address value 7'h7E
31:0	Slave address register	W	0x0	This register contains slave address value for I3C devices